

Name.....

Roll No.....



# ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE FIRST INTERNAL TEST- MARCH 2024

VIII Semester

CST 468 & BIOINFORMATICS (ELECTIVE V)

Time: 1.5 Hours

Max. Marks: 50

## PART A

Answer all questions (3 x 6 = 18 Marks)

| Qn. No. | Questions                                                                                                                                 | Marks | K level of Qn. | CO & KL  |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------|----------|
| 1       | i) Group the following as nitrogenous bases and nucleosides:<br>Adenine, Cytidine, Thymine, Guanosine, Uracil and Cytosine                | 1     | K1             | CO1&K2   |
| 1       | ii) If the sequence of one strand of DNA is written as<br>5' ATGCATGCATGCATGCATGC 3'. Write down the<br>sequence of complementary strand. | 1     | K2             | CO1&K2   |
|         | iii) If a double stranded DNA has 40% of cytosine, calculate % of<br>adenine in DNA.                                                      | 1     | K2             | CO1&K2   |
| 2       | What are the functions of mRNA, tRNA and rRNA?                                                                                            | 3     | K2             | CO1&K2   |
| 3       | Is Bioinformatics interdisciplinary? Justify your answer.                                                                                 | 3     | K2             | CO1&K2   |
| 4       | Show the visual inspection of the regions of similarity between<br>sequences using dot plot.                                              | 3     | K2             | CO2 & K3 |
| 5       | Define protein sequence database with an example                                                                                          | 3     | K1             | CO2 & K3 |
| 6       | How <b>gap penalties</b> affect sequence alignment?                                                                                       | 3     | K2             | CO2 & K3 |

## PART B

Answer any two questions (8 x 2 = 16 Marks)

Module 1: Qn. No 7,8 and 9

| Qn. No. | Questions                                                                                                                                          | Marks | K level of Qn. | CO & KL  |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------|----------|
| 7       | State and sketch the "central dogma" of molecular biology,<br>complete with all labels and indicate the responsibility of<br>enzymes in each step. | 8     | K2             | CO1 & K2 |
| 8       | With a neat sketch differentiate DNA and RNA                                                                                                       | 8     | K2             | CO1 & K2 |
| 9       | Differentiate Bioinformatics and Computational Biology.<br>Enumerate any 5 applications of bioinformatics.                                         | 8     | K2             | CO1 & K2 |

**PART C**

Answer any two questions (8 x 2 = 16 Marks)

Module 2: Qn. No 10,11 and 12

| Qn. No. | Questions                                                                                                                                                                                                                                                                            | Marks | K level of Qn. | CO & KL |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------|---------|
| 10      | Consider the sequences S1 = ATCG and S2 = TCG Assume that the match score is +1, mismatch score is 0 and gap penalty is -1. Construct the dynamic programming alignment grid for a global alignment between sequences, S1 and S2. What is the score of the optimal global alignment? | 8     | K3             | CO2 &K3 |
| 11      | Explain the importance of primary, secondary and composite databases in Bioinformatics.                                                                                                                                                                                              | 8     | K2             | CO2 &K3 |
| 12      | How global alignment differ from local alignment. What are the applications of sequence alignment in Bioinformatics?                                                                                                                                                                 | 8     | K2             | CO2 &K3 |

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**B. TECH DEGREE SECOND INTERNAL TEST- APRIL 2024**
**VIII Semester**
**CST 468 & BIOINFORMATICS (ELECTIVE V)**

Time: 1.5 Hours

Max. Marks: 50

**PART A**

Answer all questions (3 x 6 = 18 Marks)

| Qn. No. | Questions                                                                          | Marks | K level of Qn. | CO & KL |
|---------|------------------------------------------------------------------------------------|-------|----------------|---------|
| 1 ✓     | What are scoring matrices? Sketch any two.                                         | 3     | K2             | CO3&K3  |
| 2       | How multiple sequence alignment is done?                                           | 3     | K2             | CO3&K3  |
| 3 ✓     | Identify the algorithm used for local alignment. Mention the steps involved in it. | 3     | K2             | CO3&K3  |
| 4 ✓     | Sketch and define the structure of amino acid.                                     | 3     | K2             | CO4 &K3 |
| 5 ✓     | Illustrate PDB.                                                                    | 3     | K2             | CO4 &K3 |
| 6       | List any 3 protein visualization tools.                                            | 3     | K1             | CO4 &K3 |

**PART B**

Answer any two questions (8 x 2 = 16 Marks)

Module 3: Qn. No 7,8 and 9

| Qn. No. | Questions                                                                                                                                                                                                                                                                                | Marks | K level of Qn. | CO & KL  |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------|----------|
| 7 ✓     | Consider the sequences S1 = AATCGG and S2 = AACGG. Assume that the match score is +1, mismatch score is -1 and gap penalty is -2. Construct the dynamic programming alignment grid for a local alignment between sequences, S1 and S2. What is the score of the optimal local alignment? | 8     | K3             | CO3 & K3 |
| 8 ✓     | Sketch and explain Prokaryotic gene structure                                                                                                                                                                                                                                            | 8     | K2             | CO3 & K3 |
| 9       | How BLAST is used for sequence alignment ?List out its types                                                                                                                                                                                                                             | 8     | K2             | CO3 & K3 |

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**PART C**

Answer any two questions (8 x 2 = 16 Marks)

Module 4: Qn. No 10,11 and 12

| Qn. No. | Questions                                                            | Marks | K level of Qn. | CO & KL |
|---------|----------------------------------------------------------------------|-------|----------------|---------|
| 10      | Sketch and explain Ramachandran plot.                                | 8     | K2             | CO4 &K3 |
| 11      | With neat sketches illustrate hierarchies of protein structure.      | 8     | K2             | CO4 &K3 |
| 12      | Explain the experimental methods to determine the protein structure. | 8     | K2             | CO4 &K3 |

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## APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Eighth Semester B.Tech Degree (R,S) Examination May 2024 (2019 Scheme)

Course Code: CST468

Course Name: BIOINFORMATICS

Max. Marks: 100

Duration: 3 Hours

## PART A

Answer all questions, each carries 3 marks.

Marks

- 1 ✓ How proteome is related to genome? (3)
- 2 ✓ Compare mRNA, tRNA, and rRNA. (3)
- 3 ✓ What are the three interpretations of dot matrix method for sequence analysis? (3)
- 4 ✓ Differentiate constant, Linear and Affine gap penalties. (3)
- 5 ✓ What are the requirements of database searching? (3)
- 6 ✓ What are the purpose of BLASTX, BLASTP and BLASTN? (3)
- 7 ✓ Compare the different methods for determining protein three dimensional structure. (3)
- 8 ✓ How to predict protein-protein interactions based on Domain Fusion? (3)
- 9 ✓ Justify the statement – Systems Biology is Modelling. (3)
- 10 ✓ List any two factors that determine the model behaviour of a biological system. (3)

## PART B

Answer any one full question from each module, each carries 14 marks.

## Module I

- 11 a) What are the nature and scope of bioinformatics? (7)
- b) Discuss bio-molecules involved in central dogma, its structure and types. (7)

## OR

- 12 a) Illustrate the steps involved in the process of transcription. (7)
- b) What are the applications of bioinformatics? (7)

## Module II

- 13 a) Discuss the various classifications of biological databases in Bioinformatics with example. (10)
- b) Differentiate Global and Local alignment. (4)

## OR

- 14 a) Discuss the importance of Dynamic programming in Sequence Alignment. (7)
- b) Illustrate the biological sequence format "FASTA". (7)

## Module III

- 15 a) Applying Smith Waterman method construct the partial alignment score table (7)  
for the following two sequences, using the scoring parameters

match score : +1, mismatch score : -1, gap penalty : -1.

ACCGTGA  
GTGAATA

Also write down the optimal global alignment between these sequences along with the optimal score.

- (b) Illustrate the structure of a gene in Prokaryotic cell with neat sketch. (7)

OR

- 16 a) Applying Needleman and Wunsch dynamic programming method, construct the (7)  
the partial alignment score table for the following two sequences, using the  
scoring parameters match score : +1, mismatch score : -1, gap penalty : -2.

AATC

AGC-

Also write down the optimal global alignment between these sequences along with the optimal score.

- b) Illustrate the structure of a gene in Prokaryotic cell with neat sketch. (7)

## Module IV

- 17 a) Explain the format of Protein Data Bank. (7)  
b) What are the features of Protein Structural Visualization program? Mention (7)  
various styles of molecular structures produced by visualization program.

OR

- 18 a) Describe how to predict interactions among proteins. (10)  
(b) Explain the features of STRING database (4)

## Module V

- 19 a) Describe the various properties of models in system biology. (10)  
b) How does data integration help in understanding complex biological pathways (4)  
and networks?

OR

- 20 a) Illustrate on variables, Parameters, and constants in Biological systems? (9)  
b) Differentiate the system state and steady state of biological system network. (5)

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